## P0733-GEAR RATIO ERROR IN 3RD — Continued

| TEST | ACTION | APPLICABILITY |
|---|---|---|
| 2 | With the DRBIII®, read Transmission DTC's<br>If any of these DTC's are present, perform their respective tests first.<br>Are the DTC's P0944, P0715, P0720, or P1794 present also?<br><br>Yes → Refer to the Transmission category and perform the appropriate symptom. If any of these DTC's are present, they will cause a gear ratio error. Perform the test for Loss of Prime first if it is present. Perform 42RLE TRANSMISSION VERIFICATION TEST - VER 1.<br><br>No → Go To 3 | All |
| 3 | With the DRBIII®, perform the 3rd gear clutch test. Follow the instructions on the DRBIII®.<br>Increase the throttle angle, TPS Degree, to 30° for no more than a few seconds.<br>**CAUTION: Do not overheat the transmission.**<br>Did the Clutch Test pass, Input Speed remain at 0?<br><br>Yes → Go To 4<br><br>No → Go To 5 | All |
| 4 | The conditions to set this DTC are not present at this time.<br>Check the gearshift linkage adjustment.<br>Gear ratio DTC's can be set by problems in the Input and Output Speed Sensor circuits. If the vehicle passes the clutch test and still sets Rear Ratio DTC(s), check the Speed Sensors for proper operation.<br>**NOTE: Remove the Starter Relay from the PDC. This will prevent the vehicle from being started in gear.**<br>Check the speed sensor wiring and connectors for good connection, then perform a wiggle test using the Transmission Simulator, Miller tool #8333 and the Electronic Transmission Adapter kit 8333-1A.<br>This DTC can also be set under extreme temperature conditions, this is usually caused by an internal problem. Verify if the problem is only experienced under extreme hot or cold conditions.<br>With the DRBIII®, check the EATX EVENT DATA to help identify the conditions in which the DTC was set.<br>Were there any problems found?<br><br>Yes → Repair as necessary.<br>Perform 42RLE TRANSMISSION VERIFICATION TEST - VER 1.<br><br>No → Test Complete. | All |
| 5 | With the DRBIII®, read Transmission DTC's.<br>Is the DTC P0870 and/or P0871 present also?<br><br>Yes → Replace the Transmission Solenoid/Pressure Switch Assembly per the Service Information.<br>Perform 42RLE TRANSMISSION VERIFICATION TEST - VER 1.<br><br>No → Go To 6 | All |